

Final Exam - EEE3094S 2020

Information

- The exam is open book.
 - There are six questions, totalling 100 marks. You must answer all of them.
 - Submit your answers in pdf format. They may be typed or handwritten. Sketches may be hand-drawn and scanned, or constructed in a program such as PowerPoint. A .pptx file containing the plots has been provided.
 - You must explain your reasoning and show your calculations for all questions.
 - If you create or modify a plot to support your answer, ensure that you reference this graphic in your answer. You should also label the graphic with the question number. It is also helpful to annotate important features to draw attention to them.
 - Closed-loop systems are in unity feedback unless otherwise stated.
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Question 1

Consider the following block diagrams:

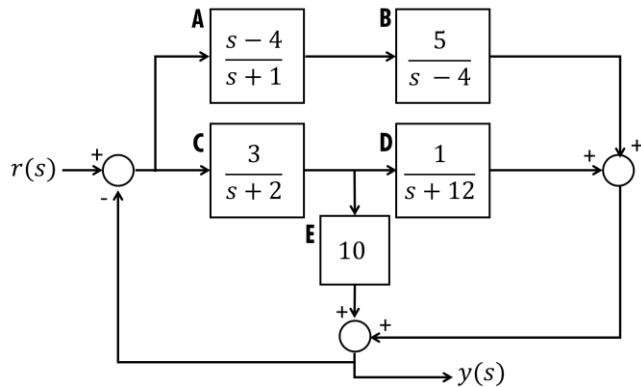


Figure 1.A - Block diagram A

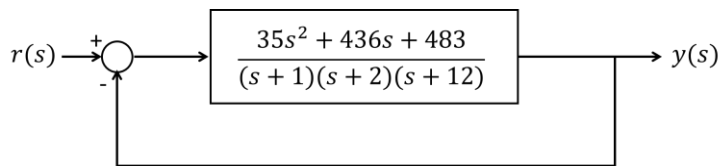


Figure 1.B - Block diagram B

- [4 marks] Show that the two diagrams are equivalent.
- [2 marks] Reduce the open-loop transfer function in the second diagram to a second-order model.
- [2 marks] Subsystem B in the first block diagram is unstable. Do you trust the stability of the overall system? Explain.

Consider the following system:

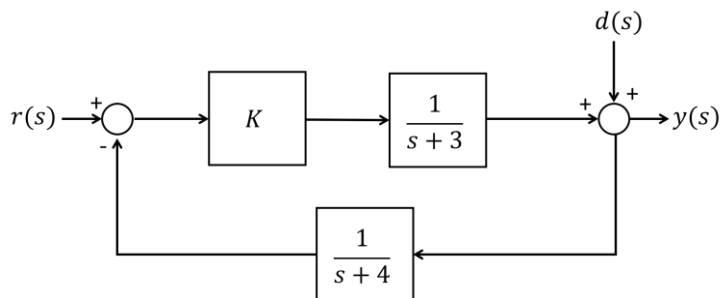


Figure 1.C - Block diagram C

- [3 marks] Taking the gain $K = 1$, find the transfer function relating the disturbance to the output of the system, and calculate its final value for a disturbance that can be modelled as $d(t) = 5u(t)$.
- [2 marks] Calculate the gain K required to reduce the effect of the disturbance to 10% of the value you worked out in (D).
- [2 marks] With reference to the type number of the system, explain how you could reduce the effect of the disturbance to zero.

Total: 15 marks

Question 2

A closed-loop system and its root locus are shown below.

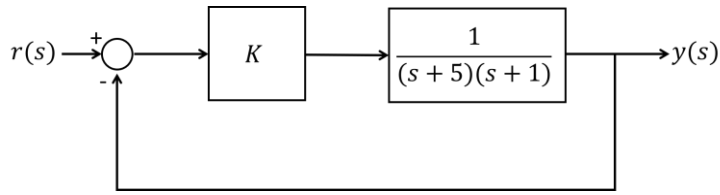


Figure 2.A – block diagram

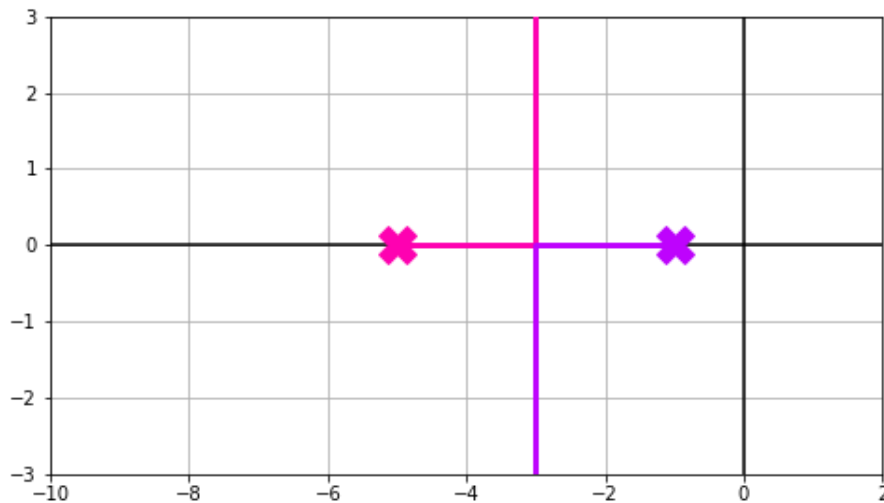


Figure 2.B – Root locus

- A) [2 marks] Calculate the gain K that places the closed-loop poles at $-3 \pm j2$
- B) [3 marks] For the closed-loop pole locations in (A), give the (i) settling time to 5% of the final value (ii) damping ratio (iii) natural frequency (in rad/s)
- C) The system experiences a time delay of $T = 0.2$ seconds.
 - i) [5 marks] Use a first-order Pade approximation to model the dead time, and draw the new root locus for the system. You must show the general shape and real axis segments of the loci, and, if applicable, the centroid and asymptotes.
 - ii) [2 marks] Does the dead time affect the stability of the system? Explain.
 - iii) [3 marks] The dead time model increases the order of the system. Would you recommend approximating the transfer function using a reduced-order model? If so, which poles are dominant, and will the model hold at all gains?

Total: 15 marks

Question 3

Decide whether each of the following statements are TRUE or FALSE. Explain your reasoning.

- A) [3 marks] The Nyquist plot in figure 3.A shows the frequency response of a system $G(s)$. This system is unstable in open loop but stable closed loop.

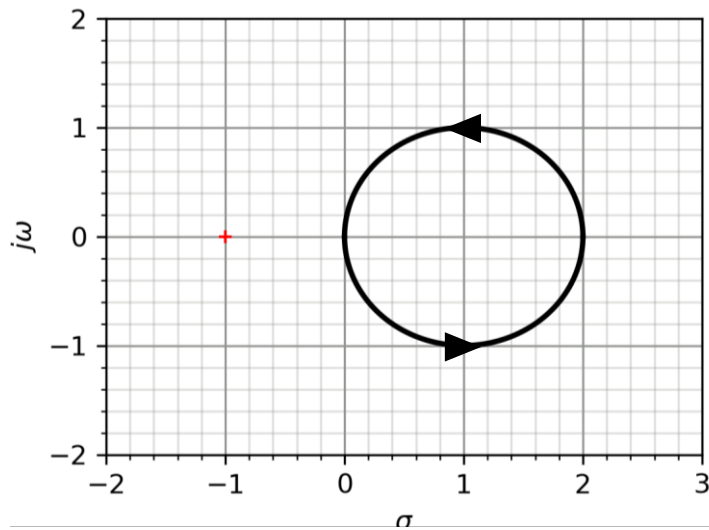


Figure 3.A Nyquist plot of $G(s)$

- B) [3 marks] The unit step response in figure 3.B was produced by the system $H(s) = \frac{1.5}{s+2}$

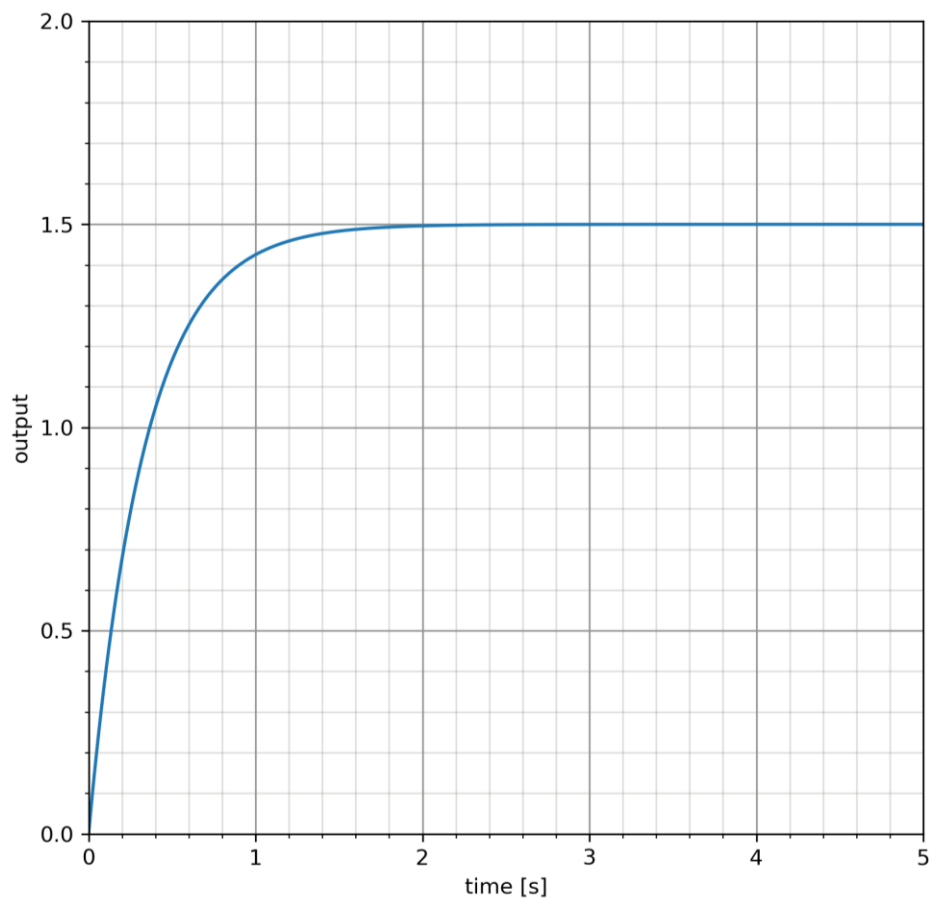


Figure 3.B unit step response

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C) [3 marks] Derivative controllers can make a system more vulnerable to noise.

D) [3 marks] A controller of the form $C(s) = \frac{K}{s}$ will enable the system $\frac{1}{s+3}$ to track a step without error.

E) [3 marks] A controller of the form $C(s) = \frac{K}{s^2}$ will enable the system $\frac{1}{s+3}$ to track a ramp without error.

Hint: you may want to support your answer to part E by sketching a rough root locus.

Question 4

Consider the system $G(s)$ with the following open-loop frequency response.

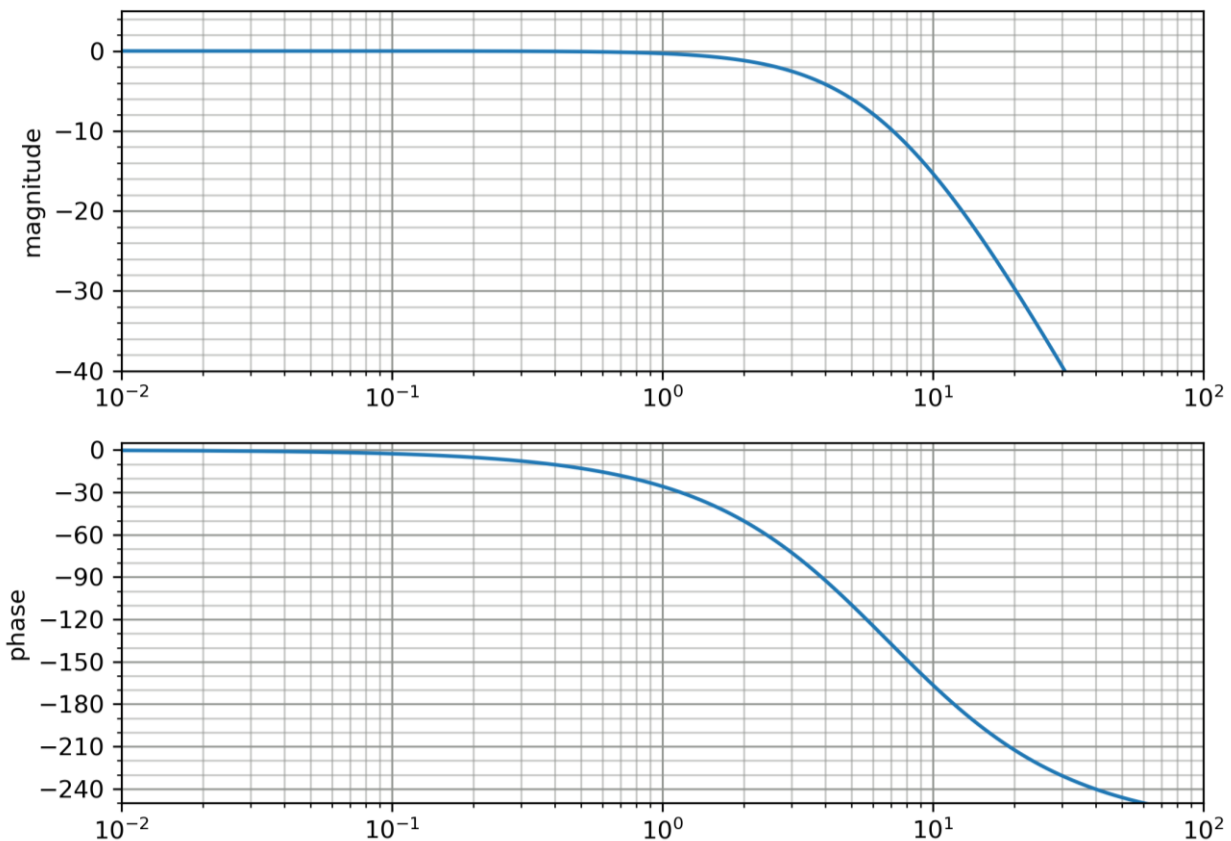


Figure 4.A: Bode plot of $G(s)$

A) [5 marks] Figure 4.B provides a template for an inverse Nichols chart. Sketch the frequency response shown in figure 4.A on that chart, labelling the points that correspond to the following frequencies:

$$\omega = [0.1, 1, 4, 10, 20]$$

B)

- [1 mark] What is the magnitude of the sensitivity function at 4 rad/s?
- [2 marks] The open-loop gain is known to vary within 5% of its nominal value due to humidity. Do you expect the closed-loop gain to vary more or less than this at 4 rad/s?

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- C) You determine that the DC gain needs to be increased by 10 decibels to achieve sufficient tracking accuracy. Your colleagues present two different designs for feedback controllers that could be used to meet this requirement:



Godzilla

$$C_2(s) = \frac{3.16(s + 1)}{10s + 1}$$

$$C_1(s) = 3.16$$

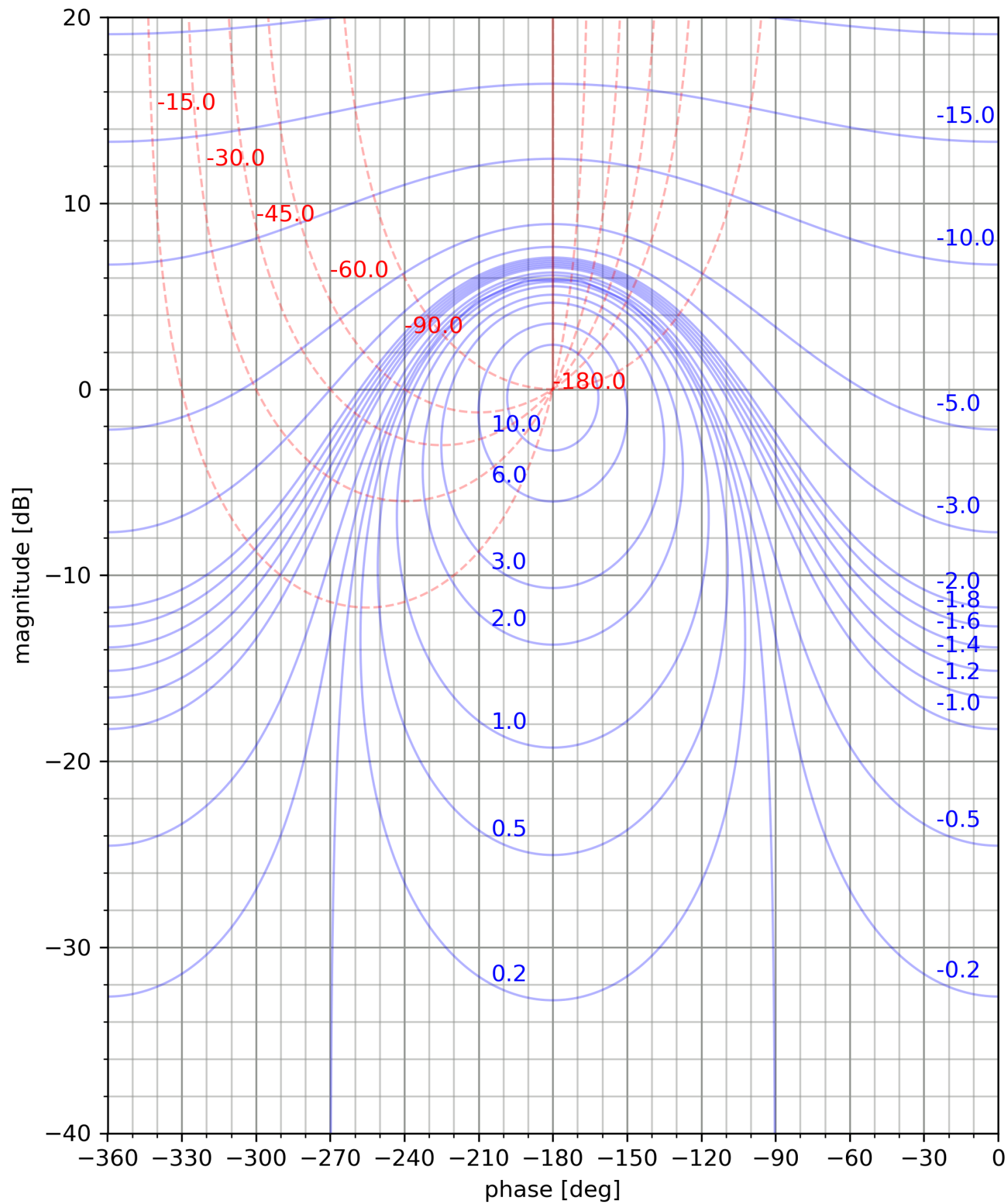


Kong

- i) [6 marks] Sketch approximate Bode plots for each design using the Bode plot templates provided in Figure 4.C. Only draw the controller, not the combined system.
- ii) [6 marks] Who do you agree with? Explain why you believe their design is superior and/ or why the other design should not be used. You may support your explanation by adding sketches to your inverse Nichols chart indicating how each design will alter the response.

Total: 20 marks

Figure 4.B: inverse Nichols chart template.



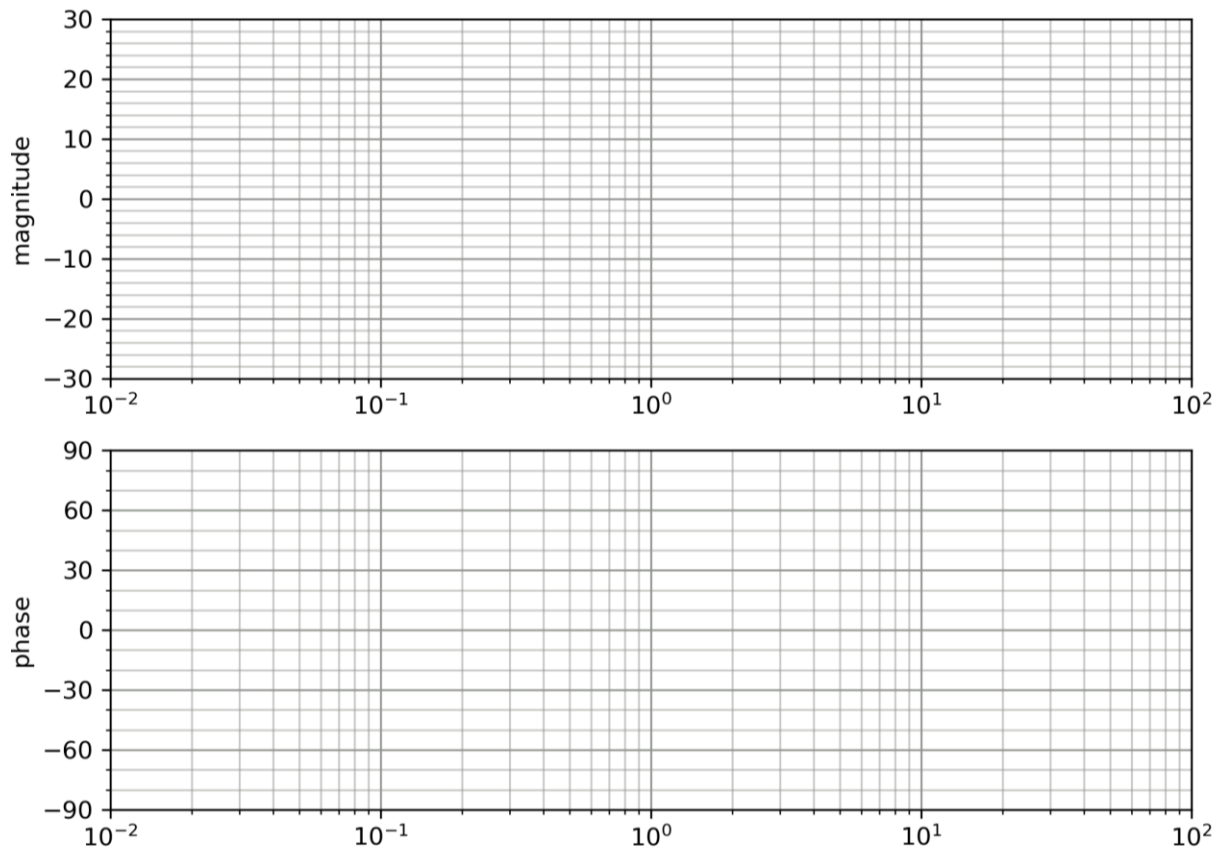


Figure 4.Ci: Frequency response of Godzilla's controller

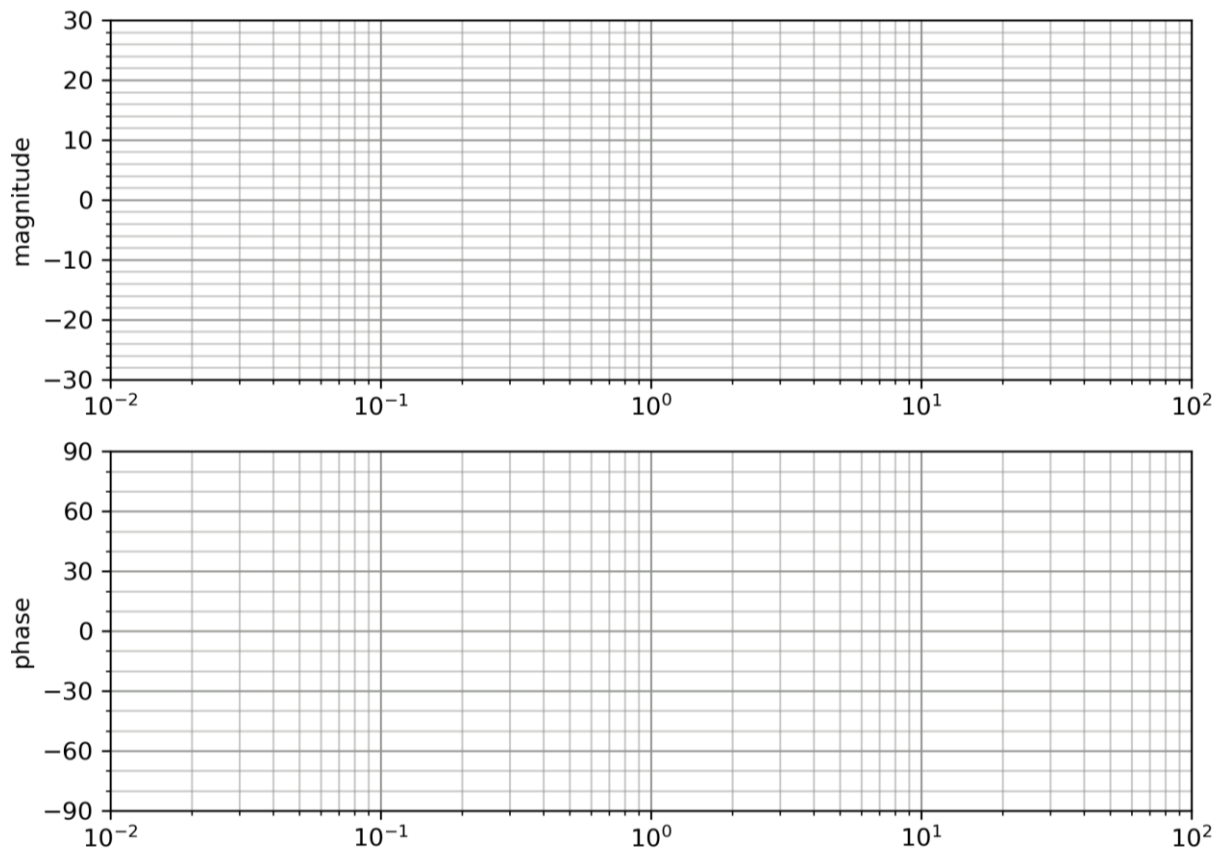


Figure 4.Cii: Frequency response of Kong's controller

Question 5:

Figures 5A-C depict the frequency response of a second-order system which is stable in open loop.

- A) The system is also stable in closed loop. Explain how you would know this by looking at each of the plots: (i) 5.A (ii) 5.B (iii) 5.C [3 marks]
- B) [2 marks] Will the system remain stable in closed loop if you increase the gain? Explain.
- C) The given plots present the results of a test to evaluate the success of a controller design for the system. For each of the following performance specifications, decide whether the system meets the requirements and explain your answers based on what you observe in these tests. You may annotate the plots to support your answers.
- (i) [3 marks] Steady-state error $\leq 5\%$ for a position input
 - (ii) [3 marks] Damping ratio of > 0.5
 - (iii) [6 marks] Settles to within 2% of final value in less than 2 seconds. (If you were unable to calculate parameters that you require for this question in previous parts, you may substitute variables or assumed values and explain how you would calculate the settling time.)
- D) Regardless of your evaluation in B, it is decided that the performance of this controller is acceptable. It is going to be implemented digitally.
- i) [1 mark] Briefly explain how sampling the system is likely to affect its frequency response.
 - ii) [2 marks] With reference to your answer in (i), would you recommend implementing the design digitally in its present form? Explain.

Total: 20 marks

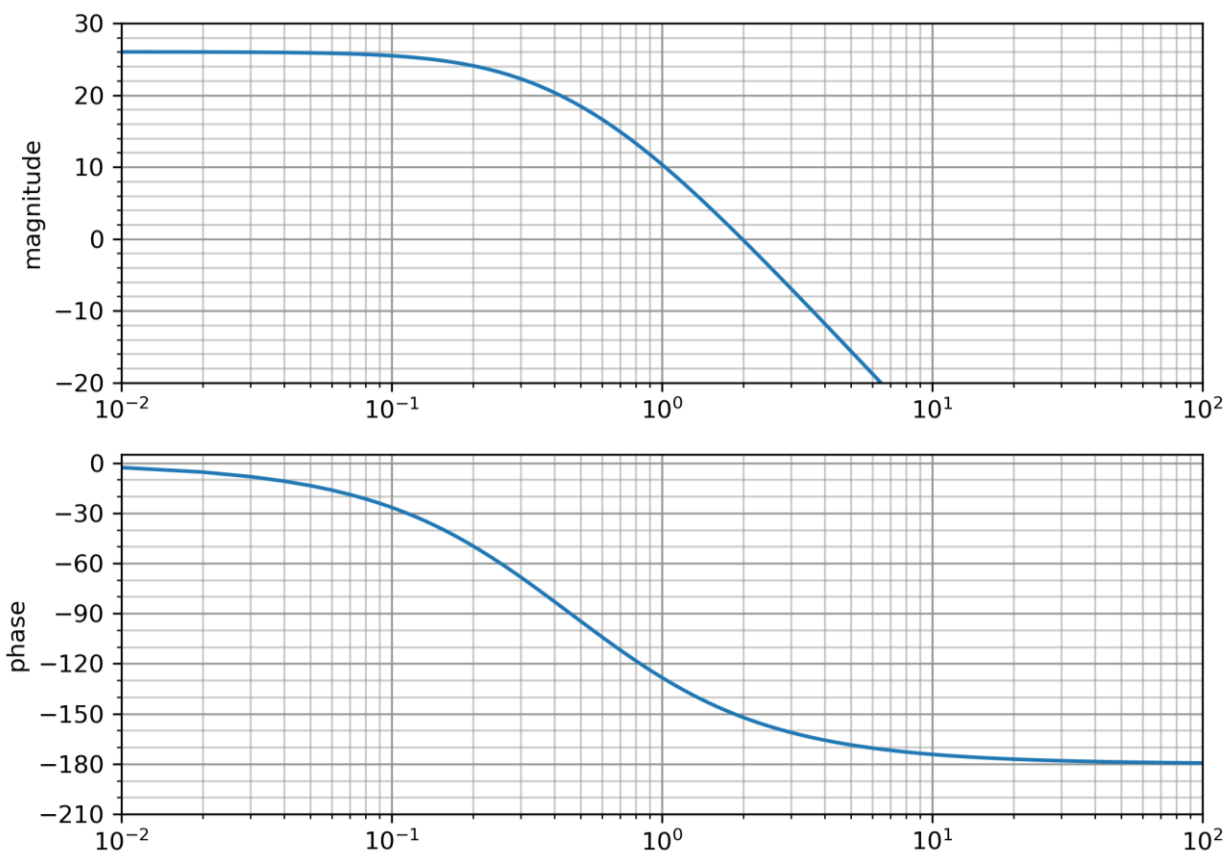


Figure 5.A: Bode plot of system

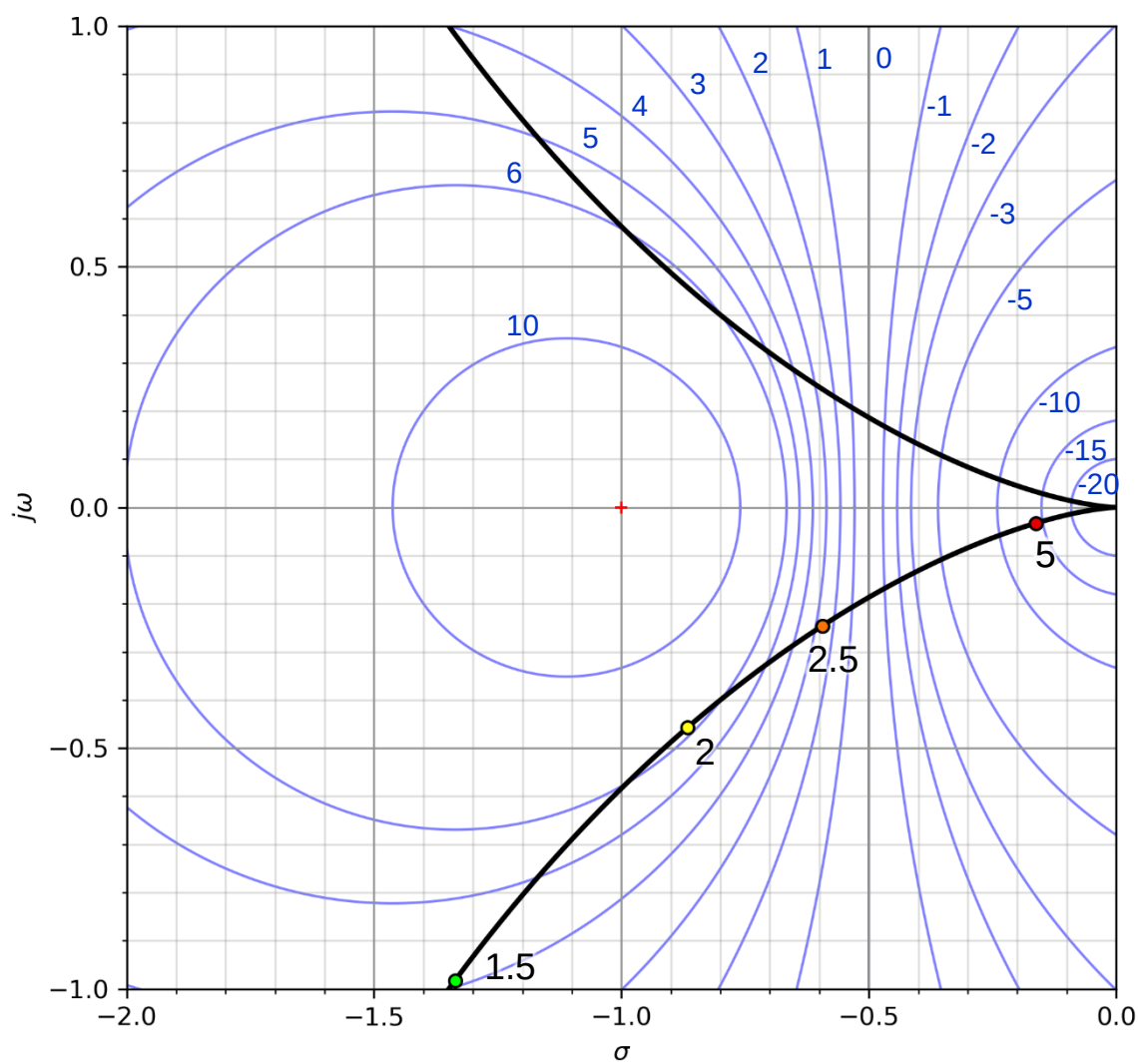
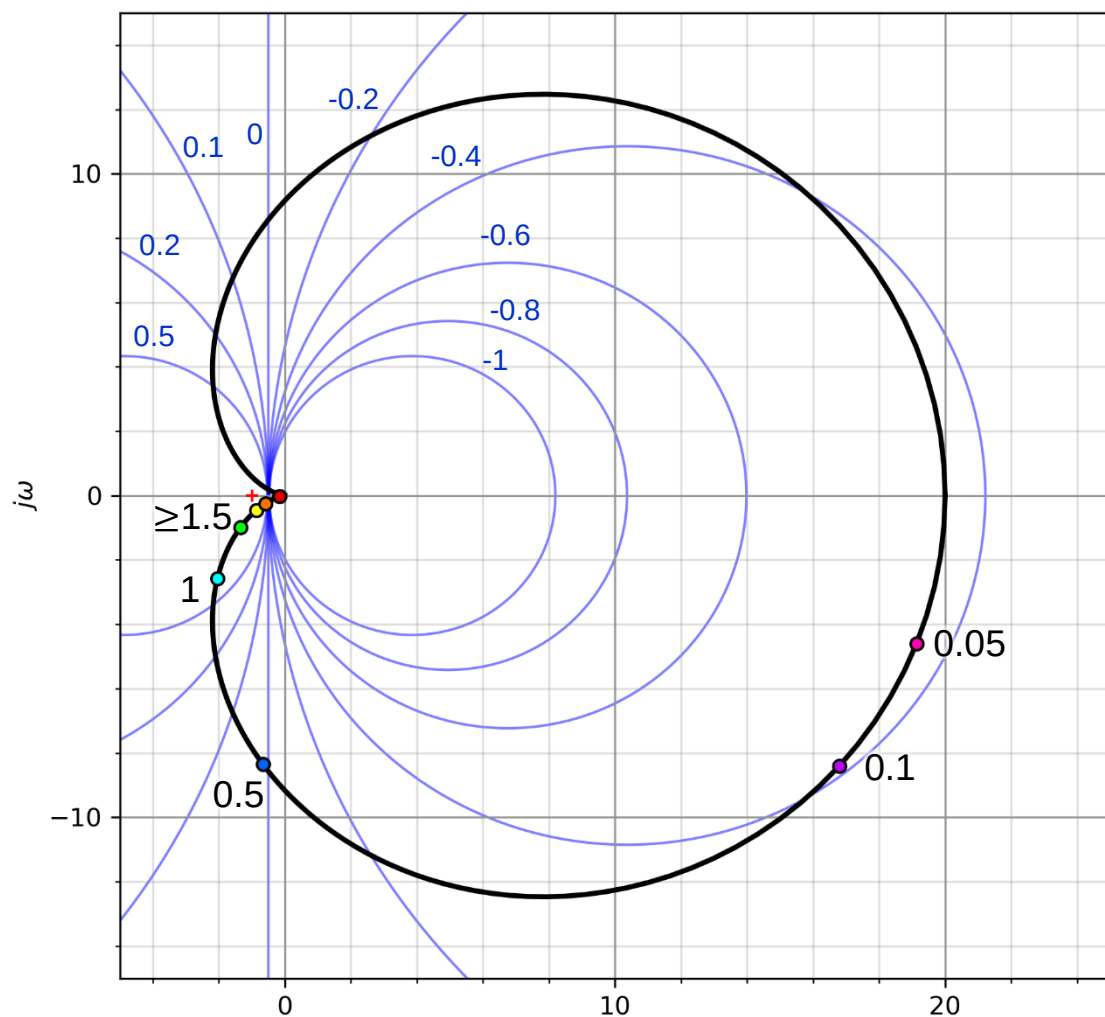


Figure 5.B Nyquist plot of system, and detailed view of Nyquist plot.

M circle magnitudes are in decibels.

Annotated frequencies are in rad/s

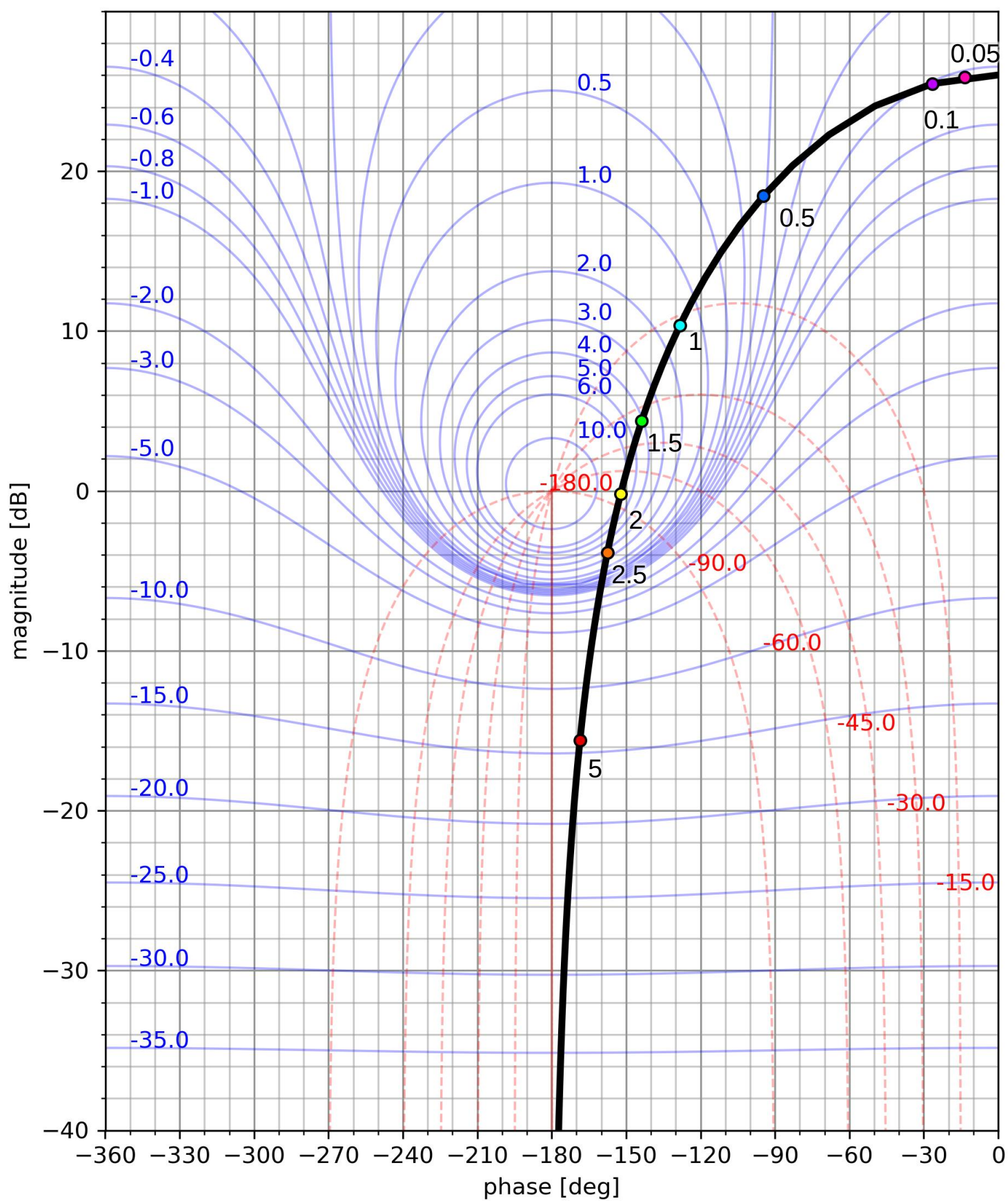


Figure 5.C Nichols chart of system. Annotated frequencies are in rad/s.

Question 6

Consider the unstable system: $G(s) = \frac{1}{(s+1)(s-1)}$

- A) Theoretically, you can stabilize this system by placing it in feedback with a controller of the form $C(s) = K(s - z)$
- [1 marks] For what range of zero positions z is it possible to stabilize the system?
 - [2 marks] If the zero is at -0.5 , find the minimum gain K for which the response will be stable.
 - [2 marks] Would it be possible to implement this controller in reality? Explain.
- B) [10 marks] Design a lead compensator to stabilize the system. The closed-loop system must satisfy the following performance requirements:
- 10% accuracy with respect to position setpoints
 - Time to peak of less than 1 second
 - Overshoot of less than 25%
 - Phase margin of at least 60 degrees to allow for sampling

Ordinarily, a design might involve multiple iterations. If you reach the end of the process and find that it cannot satisfy the specifications based on some value you selected, you do not have to continue: you can just give a short recommendation for how the design should be modified.

Formulas:

Percentage overshoot: $100e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$ Peak time: $\frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$

where ζ is the damping ratio and ω_n is the natural frequency.

Total: 15 marks

